Chapter 1

Number Sequences

Skills: investigate, model, represent, calculate, communicate

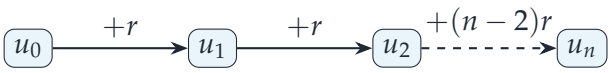
1.1 Core Course Notes

A numerical sequence assigns to each natural number n a number u_n. The index n denotes a position in the sequence; it must not be confused with the value u_n.

1.1.1 Arithmetic sequence

Key points

A sequence (u_n) is **arithmetic** when there is a real number r such that, for all n, $u_{n+1} = u_n + r$. The number r is the common difference. If the first term is u_0, then $u_n = u_0 + nr$. If the first term is u_1, then $u_n = u_1 + (n-1)r$.



Three numbers x, y, z, written in this order, are three consecutive terms of an arithmetic sequence if $y - x = z - y$, that is, if $y = \dfrac{x+z}{2}$. This last quantity is the arithmetic mean of x and z.

Method

To model an additive situation: 1) define u_n and state the unit; 2) identify the first term; 3) express the constant increase or decrease as the common difference r; 4) write the general term; 5) answer with a sentence in context.

Worked example

A shop sells 480 units in the first month and then 25 more units each month. With $u_1 = 480$ and $r = 25$, we get $u_n = 480 + 25(n-1)$. Thus $u_{12} = 755$: the shop expects to sell 755 units in the twelfth month.

1.1.2 Sum of arithmetic terms

For the $n+1$ terms from u_0 to u_n,

$$u_0 + u_1 + \cdots + u_n = (n+1)\frac{u_0 + u_n}{2}.$$

The formula is easy to remember as: *number of terms multiplied by the mean of the first and last terms.*

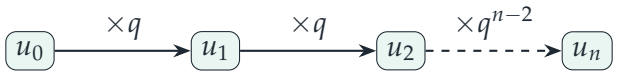
Common mistake

In $u_3 + u_4 + \cdots + u_{12}$, there are $12 - 3 + 1 = 10$ terms, not 9 or 12.

1.1.3 Geometric sequence

Key points

A sequence (u_n) is **geometric** when there is a real number q such that $u_{n+1} = qu_n$. If the first term is u_0, then $u_n = u_0 q^n$. If the first term is u_1, then $u_n = u_1 q^{n-1}$.



An increase of t % corresponds to $q = 1 + \dfrac{t}{100}$; a decrease of t % to $q = 1 - \dfrac{t}{100}$. Three positive terms x, y, z are consecutive in a geometric sequence if $y^2 = xz$; $y = \sqrt{xz}$ is their geometric mean.

For $q \neq 1$,

$$u_0 + u_1 + \cdots + u_n = u_0 \frac{1 - q^{n+1}}{1 - q}.$$

1.1.4 Algorithms: computing a sum

```
def somme_carres(n):
    s = 0
    for k in range(1, n + 1):
        s = s + k**2
    return s
```

The accumulator `s` holds the partial sum; the counter `k` runs through the integers from 1 to n.

Session 1 — Arithmetic sequences: recognising and modelling

Number Sequences

15 Sep 2026 • B: 2 h – C: 1 h

Session objectives

- Recognise an additive change.
- Find the common difference.
- Define a sequence in context.

Fluency – 6 to 8 minutes, no calculator

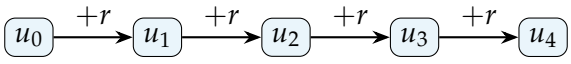
1. Write the multiplier for a 10 % increase. .....
2. Calculate u_4 if $u_0 = 14$ and $u_{n+1} = u_n + 3$. .....
3. How many terms from u_3 to u_{15}? .....
4. Give the multiplier for a 8 % decrease. .....
5. Give the common difference of $12, 12+4, 12+2\times 4, \ldots$ .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: recognise an additive change and find the common difference.

Investigation

A club has 320 members. It expects to gain 18 more each year. Let u_n be the number of members after n years. Calculate u_0, u_1, u_2. Which operation takes you from one term to the next? Write a relation between u_{n+1} and u_n, then predict u_5.



Written record

A sequence is arithmetic when $u_{n+1} - u_n = r$ is constant. In a real-life situation, a constant absolute increase leads to an arithmetic sequence.

Worked example

Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. The sequence (u_n) satisfies $u_0 = 110$ et $u_{n+1} = u_n + 3$. Write u_n then calculate u_{12}.
2. A quantity is worth 420 and decreases by 3 % per period. Write v_n and calculate v_5.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

Textbook references

Further reading: Calao, course notes pp. 18–21; exercises p. 24, no. 1–3. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 24, exercise 5

For each sequence, state whether it is arithmetic and, if so, give its common difference.

a. For every integer n: $u_n = 2(n-5) + 3n$.

b. For every integer n: $v_n = n^2 + n - 1$.

c. For every integer n: $w_n = -4 + 3n$.

Homework

Redo practice exercise 2 with the following data: replace the initial value by 125 and the rate or common difference/ratio by 3. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 2 — Arithmetic sequences: the general term

Number Sequences

17 Sep 2026 • B: 1 h – C: 2 h

Session objectives

- Move from the recurrence relation to the general term.
- Calculate a distant term.
- Interpret the index.

Fluency – 6 to 8 minutes, no calculator

1. Simplify $3x + 5 - x + 2$. .....
2. How many terms from u_3 to u_{15}? .....
3. Give the multiplier for a 8 % decrease. .....
4. Give the common difference of $12, 12 + 5, 12 + 2 \times 5, \ldots$ .....
5. Calculate u_4 if $u_0 = 15$ and $u_{n+1} = u_n + 3$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: move from the recurrence relation to the general term and calculate a distant term.

Investigation

A shop has 1 250 active loyalty cards at index 0 and gains 75 per month. Without calculating all the terms, find the number of cards at index n, then at index 18. Explain the role of 18 in the calculation.

Written record

If (u_n) is arithmetic with common difference r, then $u_n = u_0 + nr$ or $u_n = u_1 + (n-1)r$, depending on the index of the first term.

Worked example

Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. The sequence (u_n) satisfies $u_0 = 120$ et $u_{n+1} = u_n + 4$. Write u_n then calculate u_{12}.
2. A quantity is worth 440 and decreases by 4 % per period. Write v_n and calculate v_5.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

Textbook references

Further reading: Calao, course notes pp. 18–21; exercises p. 24, no. 5–8. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 24, exercise 6

Let (l_n) be the arithmetic sequence with common difference 2 and first term $l_0 = -1$.

a. Express l_n for every natural number n.

b. Calculate l_{35} and l_{64}.

Homework

Redo practice exercise 3 with the following data: replace the initial value by 130 and the rate or common difference/ratio by 4. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 3 — Sum of the terms of an arithmetic sequence

Number Sequences

22 Sep 2026 • B: 2 h – C: 1 h

Session objectives

- Count the terms of a sum.
- Apply the formula.
- Check the order of magnitude.

Fluency – 6 to 8 minutes, no calculator

1. Calculate 20 % of 120. .....
2. Give the multiplier for a 8 % decrease. .....
3. Give the common difference of $12, 12+6, 12+2\times 6, \ldots$ .....
4. Calculate u_4 if $u_0 = 16$ and $u_{n+1} = u_n + 3$. .....
5. How many terms from u_3 to u_{15}? .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: count the terms of a sum and apply the formula.

Investigation

A company pays a monthly bonus: 45 euros in the first month, then 5 euros more each month. Calculate the total paid over the first twelve months in three ways: explicit sum, pairing first and last terms, formula.

Written record

A sum of arithmetic terms equals number of terms $\times \dfrac{\text{first} + \text{last}}{2}$. From u_p to u_q, there are $q - p + 1$ terms.

Worked example

Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. The sequence (u_n) satisfies $u_0 = 130$ et $u_{n+1} = u_n + 5$. Write u_n then calculate u_{12}.
2. A quantity is worth 460 and decreases by 5 % per period. Write v_n and calculate v_5.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

Textbook references

Further reading: Calao, course notes pp. 20–21; exercises p. 24, no. 11–13, then p. 28, no. 43–44. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 24, exercise 9

a. Calculate the sum of the first 500 nonzero integers $\sum_{i=1}^{500} i$.

b. Calculate the sum of the integers from 35 to 150: $\sum_{k=35}^{150} k$.

Homework

Redo practice exercise 2 with the following data: replace the initial value by 135 and the rate or common difference/ratio by 5. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 4 — Geometric sequences: recognising and modelling

Number Sequences

24 Sep 2026 • B: 1 h – C: 2 h

Session objectives

- Recognise a multiplicative change.
- Convert a rate into a multiplier.
- Compare arithmetic and geometric sequences.

Fluency – 6 to 8 minutes, no calculator

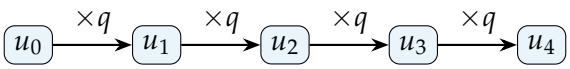
1. Write the multiplier for a 25 % increase. .....
2. Give the common difference of $12, 12+7, 12+2\times 7, \ldots$ .....
3. Calculate u_4 if $u_0 = 17$ and $u_{n+1} = u_n + 3$. .....
4. How many terms from u_3 to u_{15}? .....
5. Give the multiplier for a 8 % decrease. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: recognise a multiplicative change and convert a rate into a multiplier.

Investigation

An online shop increases its number of visitors by 6 % per month, starting from 8 000 visitors. Calculate the first three values. Compare the differences, then the quotients. Identify the type of sequence.



Written record

A sequence is geometric when $\dfrac{u_{n+1}}{u_n} = q$ is constant (non-zero terms). A constant relative change leads to a geometric sequence.

Worked example

Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. The sequence (u_n) satisfies $u_0 = 140$ et $u_{n+1} = u_n + 6$. Write u_n then calculate u_{12}.
2. A quantity is worth 480 and decreases by 6 % per period. Write v_n and calculate v_5.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a

difference or a quotient.

Textbook references

Further reading: Calao, course notes pp. 23–25; exercises p. 25, no. 14–18. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 25, exercise 19

Let (l_n) be the geometric sequence with common ratio 2 and first term $l_0 = -1$.

a. Express l_n for every natural number n.

b. Calculate l_{12} and l_{20}.

Homework

Redo practice exercise 3 with the following data: replace the initial value by 140 and the rate or common difference/ratio by 6. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 5 — Geometric sequences: general term and sum

Number Sequences

29 Sep 2026 • B: 2 h – C: 1 h

Session objectives

- Write the general term.
- Calculate a geometric sum.
- Interpret a cumulative change.

Fluency – 6 to 8 minutes, no calculator

1. Simplify $3x + 8 - x + 2$. .....
2. Calculate u_4 if $u_0 = 18$ and $u_{n+1} = u_n + 3$. .....
3. How many terms from u_3 to u_{15}? .....
4. Give the multiplier for a 8 % decrease. .....
5. Give the common difference of $12, 12 + 8, 12 + 2 \times 8, \ldots$ .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: write the general term and calculate a geometric sum.

Investigation

A capital of 250 000 Ar is invested at 4 % per year, compound interest. Write the capital C_n after n years. Calculate C_6, then the sum $C_0 + \cdots + C_6$, and explain what this last sum represents mathematically and why it is not the final capital.

Written record

If (u_n) is geometric with common ratio q, $u_n = u_0 q^n$ ou $u_n = u_1 q^{n-1}$. For $q \neq 1$, $u_0 + \cdots + u_n = u_0 \dfrac{1 - q^{n+1}}{1 - q}$.

Worked example

Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. The sequence (u_n) satisfies $u_0 = 150$ et $u_{n+1} = u_n + 2$. Write u_n then calculate u_{12}.
2. A quantity is worth 500 and decreases by 2 % per period. Write v_n and calculate v_5.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

Textbook references

Further reading: Calao, course notes pp. 25–28; exercises p. 25, no. 19–25, then p. 28, no. 42–44. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 25, exercise 22

Consider the geometric sequence (u_n) with common ratio 3 and first term $u_0 = 5$.

a. Express u_n for every natural number n. Deduce u_7.

b. Calculate: $S = \sum_{i=0}^{7} u_i = u_0 + u_1 + \cdots + u_7$.

Homework

Redo practice exercise 2 with the following data: replace the initial value by 145 and the rate or common difference/ratio by 7. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 6 — Review and formative assessment

Number Sequences

1 Oct 2026 • B: 1 h – C: 2 h

Session objectives

- Choose the right model.
- Use the formulas.
- Analyse one's own mistakes.

Fluency – 6 to 8 minutes, no calculator

1. Calculate 5 % of 180. .....
2. How many terms from u_3 to u_{15}? .....
3. Give the multiplier for a 8 % decrease. .....
4. Give the common difference of $12, 12+9, 12+2\times 9, \ldots$ .....
5. Calculate u_4 if $u_0 = 19$ and $u_{n+1} = u_n + 3$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: choose the right model and use the formulas.

Investigation

Sort eight situations into three categories: arithmetic, geometric, neither. For each situation, underline the word or piece of data that justifies the choice, then choose the appropriate formula.

Written record

Before calculating, identify: type of sequence, first index, common difference or ratio, and the term or sum asked for. The conclusion states the unit and the index.

Worked example

Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. The sequence (u_n) satisfies $u_0 = 160$ et $u_{n+1} = u_n + 3$. Write u_n then calculate u_{12}.
2. A quantity is worth 520 and decreases by 3 % per period. Write v_n and calculate v_5.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

Textbook references

Further reading: Calao, p. 29, no. 49–52. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of

the textbook.

Calao textbook – p. 29, exercise 49

For each statement, say whether it is true or false. Justify your answer.

1. A capital of 5 000 € is placed in an account with simple interest at an annual interest rate of 4 %. This situation can be modelled by a geometric sequence with common ratio 1.04.
2. The number of sparrows falls by 20 % per year in a region. An association decides to reintroduce 200 sparrows each year. This situation can be modelled by an arithmetic sequence with common difference 200.
3. The mass of household waste falls by 10 % per year. This situation can be modelled by a geometric sequence with common ratio 1.1.
4. A company's output increases every week by 200 items produced. This situation can be modelled by an arithmetic sequence with common difference 200.
5. The number of subscribers of a YouTuber increases by 15 % every month. This situation can be modelled by an arithmetic sequence with common difference 15.
6. The annual rent of an apartment rises by 320 € every year. This situation can be modelled by an arithmetic sequence with common difference 320.
7. On 1st January of a given year, Enzo places 1 000 € with compound interest in an account whose annual rate is 2 %, and on 1st January of each following year he adds 200 € to this account. This situation can be modelled by a geometric sequence with common ratio 1.02.

Homework

Redo practice exercise 3 with the following data: replace the initial value by 150 and the rate or common difference/ratio by 2. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.